SOLPS-ITER v. 3.0.9 and 3.1.1 Release Notes

March 25th, 2024

These Release Notes give a summary list of the changes from SOLPS-ITER versions 3.0.8/3.1.0 to 3.0.9/3.1.1 and give information about how to transition an older SOLPS-ITER run to be continued with the 3.0.9 or 3.1.1 code versions. Users are encouraged to read this document carefully and thoroughly before proceeding with use of SOLPS-ITER 3.0.9 or 3.1.1. Users should also consult the updated version of the SOLPS-ITER Manual that accompanies this distribution and other documents posted to the SOLPS-ITER web site and Confluence page. In particular, the minutes from the SOLPS-ITER User Forums will provide more detailed information about the features discussed below. For full details of all updates, please consult the SOLPS-ITER Manual and GIT repository logs. In addition to the items listed below, the new code version also contains lots of bug fixes, new documentation, updates to the SOLPS-ITER Manual and configuration files, code clean-up, and output clarification/prettifying improvements, as reported/requested by users and developers since the release of the 3.0.8 and 3.1.0 versions. Some of the changes that have occurred on the **Eirene** side are documented in the updated **Eirene** manual, which is now part of the code distribution.

Note: The 3.0.9 and 3.1.1 code versions differ mostly in the numerical scheme used for solving the plasma transport equations. The 3.0.9 version relies on a 5-point stencil, while the 3.1.1 version allows for use of a 9-point stencil (with backward compatibility to the 5-point stencil solution method). They are otherwise nearly identical in terms of features and physics model. Notably, they couple to the same **Eirene** version. For the remainder of this document, unless explicitly stated, the 3.0.9 moniker applies to both the 3.0.9 and 3.1.1 code versions. These release notes do not apply to the Wide Grids 3.2.0 SOLPS-ITER code version.

If you are transitioning from SOLPS-ITER v.3.0.6 or earlier to v.3.0.9, then you should also consult the v.3.0.7 and v.3.0.8 Release Notes and their addenda (here, here, and here).

The code is freely distributed upon request to civilian state and academic institutions within the ITER Member States, provided they have signed an IMAS Cooperation Agreement (CA) with the ITER Organization for that purpose. You can consult the list of signatory institutions here. Other entities may also sign an IMAS CA, subject to approval by the ITER Legal Office. Signatory institutions have the option to sponsor third party users. The code is strictly licensed for non-commercial purposes only. You will find the SOLPS-ITER `LICENSE` document in the top directory of the code distribution, and as an Appendix to the SOLPS-ITER Manual. There is an additional `LICENSE` document in each of the **B2.5**, **Carre**, and **DivGeo** submodules which covers their usage as stand-alone programs. Users requiring access to the **Eirene** repository managed at ITER as part of the SOLPS-ITER package also need to abide by the **Eirene** license (found as the `EPL` file in the top **Eirene** directory), which is managed by Forchungszentrum Jülich in Germany (check www.eirene.de for more details).

Users that have not done so already are encouraged to subscribe to the SOLPS-ITER Slack channel (solps.slack.com). Simply send an email request to the IO Responsible Officer (xavier.bonnin@iter.org) to receive an invitation to join if you need one.

1. How to obtain SOLPS-ITER v.3.0.9:

In order to update your copy of the SOLPS-ITER code suite from earlier versions to 3.0.9 (3.1.1), (switch to the *rc/3.0.9* or *rc/3.1.1* SOLPS-ITER branches and) use the *solps_iter_update* or *solps-iter_update_3.0.9_release* (*solps-iter_update_3.1.1_release*) scripts. If coming from version 3.0.5 or older, run the *switch_to_master* script first. As of version 3.0.6, the SOLPS-ITER distribution does, by default, point to the *master* branch of the SOLPS-ITER GIT repository and its submodules. If you wish to point instead to the *develop* branch, then use the *solps-iter_update_develop* script. As usual, before making any such update, you must make sure that any local changes you might have made are committed to local GIT branches or stashed, so as to avoid them being overwritten or creating conflicts during the update attempt, which will interrupt the process.

Users that also need the SOLPS4-5 converter program **b2sxdr** should update their copy of SOLPS4.3 as well. For the **Carre2** code, there is a new dedicated branch *feature/Carre2_3.0.9* in the **Carre** repository.

For subsequent updates, users should use *solps-iter_update* or *solpsiter_update_develop*, depending on which branch they are using, and only once per update.

IMPORTANT: Because the update to v.3.0.9 involves a large number of configuration file changes (and the switch to `CMake` when compiling Eirene), new environment modules (in particular, moving to IMAS Access Layer 5), and a significant number of added/renamed/moved/removed source code files, it is likely that the compilations launched by the update scripts will fail the first time around. If this is the case, start afresh in a new window, load the updated SOLPS-ITER environment as usual by means of the *setup.csh* file, and try again to compile. If it fails once more, then do a `gmake clean` first and compile again. Recall that you can use `gmake clean_XXX` to remove all objects associated with any `XXX` build, for example `gmake clean_solps_mpi`. Specific to this release, it will be necessary to rebuild the Triang executables from a clean state.

2. New features:

2.1. Coupling to Eirene Milestone Version

This is the most important development leading to this new SOLPS-ITER version. The **Eirene** code has undergone extensive streamlining resulting into a new single reference "Milestone version" (MsV). A detailed listing of the changes that constitute the MsV development is outside the scope of these Release Notes and users are referred to the **Eirene** Manual. A set of additional changes necessary for SOLPS-ITER coupling have been added on top of the MsV branch commits. This also means an update to the **Eirene** license, which all users wanting to use **Eirene** in their SOLPS-ITER runs must adhere to. The main features users should be aware of include:

- The **Eirene** Manual is now part of the code distribution and builds automatically as part of the compilation and can be found as *$SOLPSTOP/modules/Eirene/Manual/eirene.pdf*.

- The **Eirene** compilation now proceeds using `CMake`. In case `CMake` is not available on your system (or the `NO_CMAKE` environment variable is declared), the overall SOLPS-ITER *Makefile* will detect this and revert to the old configuration files for building the code. To allow this, the old SOLPS-ITER **Eirene** *Makefile* has been moved to the *config/* directory. You can provide the location of the graphics libraries needed by **Eirene**, which will be passed to `CMake`, by means of the `LIBGKS` and `LIBGRS` variables, which are defined according to the known environment set-ups, in the *$SOLPSTOP/SETUP/config.$HOST_NAME.$COMPILER(.local)* files. It was found on some systems that the *.so* version of the GKS library was needed instead of the *.a* one, so users should feel free to change this (using a local copy of their configuration file) and ROs on the various clusters where centralized installations are available should commit the recommended settings for their systems to the general distribution. At ITER, we also needed to link to a more modern version of the GKS library than that provided as part of the original `GR` module.
- It is now possible to use OpenMP parallelization when running **Eirene**. This parallelization is done by assigning individual strata, or a fraction of a stratum, to different threads. Compilation of **Eirene** with OpenMP options proceeds in the same way as for **B2.5**, by adding the `_openmp` suffix to the compilation target, e.g. `gmake b25eirene_openmp`.
- **Eirene** can now read its input file using the JSON format, provided a `json-fortran` library is available on the system (which `CMake` automatically looks for or can be specified explicitly by means of either the environment variable `JSONF_LIB_DIR` or the `LibJSON` argument to the `cmake` command). If one starts from a traditional formatted ASCII file, **Eirene** will translate the input file into JSON equivalents (*eirene.input.json*, *eirene_add_surfaces.input.json*, and *eirene_physics_model.input.json*), which can be used in further runs. If no JSON library is available (or the `NO_JSON` compiler flag is given), the code reverts to the traditional ASCII input.
- The **Eirene** reaction database files have been significantly updated. The SOLPS-ITER *Makefile* will make sure that links to the needed database files are created so that they are found in their usual locations.

2.2. Support for IMAS Access Layer 5:

The code can now make use of IMAS Access Layer 5 features, if available in your environment. It is also compatible with IMAS Data Dictionary (DD) versions up to 3.40.1. When setting up your list of IMAS modules, make sure that the GGD and AMNS versions match the Access Layer version: `GGD/1.10.x` and `AMNS/1.4.x` for Access Layer 4, and `GGD/1.11.x` and `AMNS/1.5.x` for Access Layer 5. The *Makefile*s automatically detect which IMAS DD, GGD, and Access Layer versions are loaded on the system and a set of compiler pre-processor pragmas in the source files build the code in accordance with the environment.

The main new feature is the ability to specify the path where the IMAS data entry containing the run results will be stored, by means of the `b2mndr_ids_path` switch in *b2mn.dat* and/or the `--path` option for the **b2_ual_write**, **b2_ual_rewrite**, **b2_ual_write_b2mod**, **dg2ids**, and **ids2dg** programs. The path can be either an absolute one (starting with a '`/`' character) or relative to the `$IMASDIR` environment variable, or can contain the `$HOME`, `$SOLPSTOP`, or `$IMASDIR` environment variables. The old options for specifying shot and run numbers, user name, database name, and DD version number can still be used to modify the default path. If no path is provided, the code looks for it as the last line in the *shotnumber.history* file, if present. If that file is not found, then the default path is set to `$IMASDIR`, if declared, or *$HOME/public/imasdb/$DEVICE/$IMAS_VERSION* otherwise. To match the IMAS database naming convention, the device name `"iter"` is automatically capitalized.

New quantities from recent versions of the Data Dictionary have been added whenever relevant. A new source array (`edge_sources%source(13)`) contains the total source (provided by **Eirene**) due to plasma-neutral interactions. If using Access Layer 5, the code only writes a single copy of the GGD specification (in the `edge_profiles` IDS) and provides a path to it in the other IDSs that use the same GGD object. Preparations have also been made in the code, by means of compiler pragmas, for DDv4 (and the associated GGD version 2.x.x), when the consequences of upcoming changes are known. The code also now makes use of the `imas-identifiers` library whenever possible. Quantities related to the toroidal direction have been corrected to match the DD sign convention, which is opposite to that used inside **B2.5**.

Note that the `--step` option in **b2_ual_write_b2mod** no longer requires an argument. If that option is selected, this simply means that the code should be run for the number of timesteps provided in the *b2mn.dat* file.

The code can now be compiled within an IMAS environment even if no GGD module is provided, but this will disable most of the IDS output and a warning message will be given at run-time. The IMAS, GGD, and AMNS version numbers being linked to can be read from the compiling instructions shown by the *Makefile*s, and users should check that they correspond to the environment they expect.

2.3. Improvements for OpenMP performance in B2.5:

Thanks to dedicated work by the Barcelona Supercomputing Center (BSC) Advanced Computing Hub, the OpenMP (and serial) performance of the **B2.5** code has been significantly improved. Bottlenecks that were limiting scaling were identified and fixed, and memory page swaps and superfluous thread creations/destructions that were slowing down the code have been avoided by recoding some utility routines. Some OpenMP environment switch settings have also been modified to improve performance and facilitate code debugging. The code now exhibits good scaling behavior (up to a number of threads commensurate with the number of species in the problem).

Other compiler options were proposed by the BSC team to improve compilation in general. In particular, the Intel compilation now tries to take advantage of local architecture optimizations, by means of the `-xHost` flag. Users that compile the code to run on multiple architectures should be aware that such optimization may not be portable, and adjust the compiler settings accordingly (as was done for the ITER SDCC cluster, for instance). These optimization flags are usually found in the *config/compiler.$HOST_NAME.$COMPILER(.local)* files.

2.4. New feedback schemes:

Several new options are now available for feedback schemes, including against the divertor neutral pressure, total SOL particle content, radiated power, target power flux density, and target peak or total saturation current. New rescaling schemes are also available, as well as the **Eirene** pumping albedo as an actuator. Please consult the code inline documentation for full details.

Note that, since negative values of `NA_FEEDBACK_IB` now refer to **Eirene** strata, the default value for this array has been changed from `-1` to `0`. One should therefore change the assignments accordingly in the *b2.feedback_control.parameters* file (or have **b2yt** do it for you).

2.5. Long-term averages:

It is now possible to produce *fort.44_aver* and *fort.46_aver* files that will contain long-term running averages of the **Eirene** output. See section 5.2 of the SOLPS-ITER Manual for details.

In addition, one can now save, as a separate occurrence of the `edge_profiles` and `edge_sources` IDSs, the batch average data from the SOLPS runs, used in the averaging scheme detailed in Section 5.6 of the SOLPS-ITER Manual.

2.6. New *b2mod_dimensions* module:

To facilitate the automated code generation process for algorithmic differentiation and uncertainty qualification, we have moved the maximum problem size definitions from the *DIMENSIONS.F* file to a *b2mod_dimensions.f* module. Variable names within remain the same, and users wanting to change the default values provided should, as before, copy this module to their *$SOLPSTOP/modules/B2.5/src.local/* directory and make any desired changes there, so that they are not overwritten or in conflict with code updates coming from the GIT repository. The *Makefile*s detect if *b2mod_dimensions.f* is present, and adapt the compilation as needed.

For the same reason, the use of statement functions and `GOTO` statements in the code has been minimized whenever possible and replaced with modularized or contained routines and `case select` statements.

2.7. New analysis plotting scripts:

Several new scripts have been provided to allow for better diagnostic of the run results, including reviving the old SOLPS4.3-style plotting scripts for large amounts of time series data, such as *b2pl*, *b2pr*, *last_2d*, *plot_series*, *plot_trc*, *plot_trg*, *sum_trg*, etc... Please consult the SOLPS-ITER manual for a full listing.

Also, there is a new *plot_trouble* script dedicated to looking at trajectories of killed particles in **Eirene**, provided its particle tracing output has been turned on.

2.8. *easybuild* installation and setup scripts:

The script *$SOLPSTOP/SETUP/easybuild-local.sh* provides a simplified way to install the code from scratch, downloading and installing the necessary support software from the IO repositories and open-source software banks. It attempts to reproduce the ITER Scientific Data Computing Center (SDCC) cluster environment. See the SOLPS-ITER Manual and the comments inside the script itself for information on how to use it.

To avoid users overwriting each other's setup when sharing a centralized installation, the re-usable environment setup files created by *setup.csh* upon first session are now saved as *$SOLPSTOP/SETUP/setup.csh.$HOST_NAME.$COMPILER.env.local.$USER*. The *$SOLPSTOP/SETUP/setup.csh.$HOST_NAME.$COMPILER.env.local* files are no longer used and can be deleted.

New configuration files have been provided for the PPPL `gfortran` environment. A new `UL` host name is recognized for the University of Ljubljana (Slovenia, EU).

It is now possible to install, compile, and run nearly all the SOLPS-ITER code suite components on MacOS, using a Darwin ARM8 architecture, and the `gfortran` compiler. The `HOST_NAME` variable will then automatically be set to `DARWIN`. Please follow the detailed instructions given in the SOLPS-ITER Manual Section 1.3.1.

2.9. Optional Manual build:

It is now possible to skip the build of the SOLPS-ITER, **DivGeo** Doxygen, and **Eirene** Manuals with the `NO_MANUAL` environment variable, for systems on which `LaTeX` is unavailable or the version too old to succeed in build the Manuals. The Manual *Makefile* also checks which `LaTeX` version is available on the system and tries to adjust the list of needed packages accordingly. Under normal conditions, the `gmake manual` command from the SOLPS-ITER top directory will build both the SOLPS-ITER and **Eirene** Manuals.

The SOLPS-ITER Manual is now available as an artefact from the CI server, and can be found here.

2.10. Job submission scripts:

The SLURM job submission scripts now have a `--no-requeue` option included to avoid a failed job being resubmitted before the user had the opportunity to look at the run output and decide on the next steps. An email is also sent to the submitter upon failure of the job.

3. Changes and extensions to the physics model:

Please be mindful that the changes to the default physics model described in this section will most likely cause some evolution of existing 3.0.8 solutions. The magnitude of that change will depend on the specifics of the cases at hand. The benchmarks and sample cases available as part of the code distribution have all been recomputed using the 3.0.9 physics model.

3.1. New minimum temperature enforcement mechanism:

Two new switches `b2upht_t[ei]_min` (default value `0.1 eV`) provide a way for the user to set the minimum electron and ion temperatures allowed in the **B2.5** solution. In previous code versions, the enforced minimum temperature was 10^{-5} eV and hard-coded.

In addition, the code now contains tests to make sure all velocity are sub-luminal, to detect cases where the solution is running away.

3.2. Extension of the Grad-Zhdanov closure scheme for bundled species:

The Grad-Zhdanov closure scheme is now available for runs with bundled species. As a reminder, its use is especially recommended when considering plasma mixtures of species with like masses, or when the trace impurity assumption does not hold. See Section C.11 of the SOLPS-ITER Manual for full details. In particular, this closure also allows for the assignment of the various currents to their contributing species.

The *ITER_Be-W_D+T+He+Ne* example case now contains a *run_Zhdanov* directory in which the Grad-Zhdanov closure has been implemented.

3.3. New option for the ion-neutral friction current:

With the availability of the perpendicular momentum sources from neutral-plasma interactions in **Eirene**, a new model for the computation of the ion-neutral friction current making use of these quantities has been implemented, activated by means of the `b2tinnt_kinetic` switch. This also allows for the assignment of this current on a per species basis.

3.4. New wall material migration model for Beryllium walls:

A new model is available to follow the erosion and migration of Beryllium from the vacuum vessel walls in coupled runs, activated as a user-defined surface model in **Eirene**. The wall elements for which this model is to be applied can be selected within **DivGeo**, and **Uinp** will create the appropriate the input files. Please consult Section 4.3 of the SOLPS-ITER manual for full details.

3.5. New pressure feedback loop scheme for Eirene:

A new recycling model has been implemented that allows for feedback control of the wall recycling coefficients in **Eirene** to maintain a certain neutral pressure in a given set of cells. See description in Section 4.2 of the SOLPS-ITER Manual. The location of the affected wall elements and regions for the pressure calculation can be selected within **DivGeo** and are indicated in blocks 3B and 6 of the **Eirene** input file as additional variables.

3.6. Support for non-standard topologies:

The code now supports snowflake(-minus) topologies, which correspond to a case with 7 volumetric regions. See Appendix H.7 of the SOLPS-ITER manual for a mapping of that topology. A standardized example is provided, along with Python scripts to help with case build-up, in *$SOLPSTOP/runs/examples/MAST_U_SFminus_Donly* .

New options have been implemented for linear device modelling, including a way to build a grid automatically, using a new `GRID` namelist stored in the *linear.geometry* file (see Section 3.1.2 of the SOLPS-ITER Manual and the code inline documentation for a full description), managed by the new *b2mod_ld_namelist* module and *b2ld* routine, and new boundary condition types for providing profiles of plasma parameters entering the domain from one side (`BCCON = 28` to `30`; `BCMOM = 18` to `21`; `BCENE = 23`; `BCENI = 28`; `BCPOT = 14`, `15`, and `17`, managed with the `b2mod_profile` module and `get_profile` routine).

A new example for limiter topologies, *$SOLPSTOP/runs/examples/JTEXT_Limiter_H*, is also available. Several important bug fixes to revive the limiter topology support are also included in the new code version.

For SN topologies in which the X-point lies on the midplane, they are identified as LSN if the X-point is to the left of the O-point, and USN otherwise.

Several analysis scripts have also been generalized to handle all the supported topologies (linear, limiter, stellarator island, and snowflake), in addition to the standard SN and DN cases.

4. Changes to the input files:

4.1. Changes to the Eirene input files:

The 3rd digit in the `NFILE` variable from block 1 of the **Eirene** input file should be set to a value larger than 5 (default recommended value is 8), to ensure that the "short" format of the *fort.13* files is being used. A new recommended default value for `NFILE` is provided by **DivGeo** and **Uinp** depending on the particular of the case at hand. Two new logical switches `NLSPCSCL` and `NLSPCSCL_ON` provide for the species-resolved particle balance rescaling scheme, which was extensively rewritten in the MsV. The `NLSOLEDGE` switch should be turned on for cases where molecular ions are expected to play an important role. Please consult the **Eirene** manual for details.

The diagnostic spectral lines provided in block 12 of the **Eirene** input files are now associated to a reaction card from block 4, see the **Eirene** Manual for a full description. Note that the treatment of line-of-sight diagnostics and the overall photon transport model have been extensively rewritten in the MsV, meaning, *inter alia*, that the `NREAC_ADD` variable from block 1 is no longer needed.

If the `NLCRR` switch is activated, asking for correlated sampling, then the value of `NINITL_READ` in block 13 for the random seed used for the census stratum will be made to be a positive number automatically. Note that the random number generator used by **Eirene** has been modernized in the MsV.

On the first line of block 14 of the **Eirene** input file, a sixth logical switch, `LOLD31`, can be used to tell **Eirene** that the *fort.31* file it will read uses the SOLPS5.3 (not SOLPS-ITER) format.

When reading the *fort.44* file, the code can now handle the case in which, from the time the file was written to when it was read, the time-dependent "census" stratum and its associated time horizon non-standard surface were removed or added. By default, the code will no longer create a census stratum, unless it is explicitly declared in block 13 of the **Eirene** input file. This will occur even if a *fort.15* file containing a non-empty census stratum is present, so an existing census stratum can be discontinued by setting `NPRNL` to `0` in block 13. Moreover, the time horizon location (set by the `DTIMV` variable in block 13) will no longer be overwritten by the `EIRENE_STEP_DT` value from *b2.neutrals.parameters*, unless that variable is present and nonzero. To facilitate this, the default value of `EIRENE_STEP_DT` was changed to `0.0`.

The format number of the *fort.44* file has been increased to 20240311. This corresponds to the addition in the header of the file of the NFLA variable from **Eirene**, i.e. the number of ion plasma background species used to write the file, which lets the code know the size of some of the arrays found later. This is helpful for cases where a case is being restarted with a new isonuclear sequence being added, so that the neutral background information no longer must be thrown away.

4.2. Artificial neutral sources speed-up scheme:

The artificial neutral sources speed-up scheme has been generalized to handle cases with multiple species, and assigns its own multiplier to each species . The switches b2stbr_sna_corr and b2stbr_taumax are thus now replaced with the SNA_CORR and TAUMAX arrays to be defined in *b2.numerics.namelist*. The **b2yt** converter program performs these changes automatically. The DO_SNA_CORR_CORE switch has also been added to specify if the user wishes the speed-up scheme not to apply in the core region.

4.3. Far SOL transport multipliers:

In *b2.numerics.parameters*, one can now specify multipliers for far SOL transport, by means of the DT[CO|MO|EE|EI]_SOL and IY_SOL_DTXX variables.

4.4. Input source profile reference locations:

In *b2.sources.profiles*, the new variables I[XY]REF_PROFILE can be used to overwrite the default values from set_transport_i[xy]ref to define the central position of the source profiles defined in the file. This can be particularly useful when defining a moving source profile as a function of time, for instance.

4.5. Time-dependent gas puffing:

New options in *b2.neutrals.parameters* can be used to specify time-varying gas-puff strengths, as a combination of linear ramps, staircase steps, and sine or triangle waves, using the TIME_DEP_PUFF_[FUNC|CASE|PARAM] variables. See Section 5.5 of the SOLPS-ITER Manual for a full description and worked out examples.

4.6. New recommended ADAS data set for Boron:

The *ratadas.filelist* file has been updated to reflect the new "year 95" recommended data set for Boron.

4.7. Miscellaneous switches:

The b2stbc_bcene_15_style switch had its default setting changed from 0 to 1, which computes the electron current directly from the sheath potential drop instead of the difference between the ion flow and the poloidal current, improving numerical stability, especially for cases with drifts. This applies to the BCENE = 15 boundary condition. The imposition of the BCENE/I = 15 boundary conditions can now be further stabilized with the new switches b2stbc_stab_coeff_sheath_t[ei]. Similarly, the BCCON=10 boundary condition can be further stabilized by the use of a positive value in the associated CONPAR(,,2) parameter.

When asking for output at regular plasma time intervals (using b2mndr_trantim, b2mndr_cdfmovietim, b2mndr_plasmatim, and/or b2mndr_ids_time), the time interval requested must be at least as large as the simulation time-step, which simplifies the output management. The run average of the balance quantities (obtained with the balance_average switch) is now computed over the number of time-steps actually executed, not the number requested by b2mndr_ntim.

When converting a case using **b2yt**, a new eirene_format value ("msv") is available to indicate that the **Eirene** input file to be converted corresponds to the (ASCII) format from the **Eirene** MsV.

The MAXPOIN variable in *b2.neutrals.parameters* is now obsolete. **Eirene** will automatically resize the contour arrays if needed. The arrays for correspondence between **Eirene** and **B2.5** species (eb2atcr, latmscl, lmolscl, and lionscl) now undergo additional checks to make sure their values remain within their valid ranges.

A new variable TE_DET_THRESHOLD (default value 2.0 eV) can be used in *b2.user.parameters* to define the part of the targets that are in detached regime for the purpose of the quantities computed in the new *user_SPb.trc* tracing file.

5. Changes to the output:

5.1. Eirene perpendicular momentum sources:

Eirene now calculates and provides the momentum sources due to neutral-plasma interactions in all three directions: parallel, radial, and diamagnetic. Within **Eirene**, the momentum tallies have changed their species dimensions from NPLS to 3*NPLS, where the first NPLS elements are for the parallel direction, the second NPLS set for the radial direction, and the third NPLS set for the diamagnetic direction. There are also additional magnetic field arrays to store these directions. To avoid confusion with the old variable names, the new vectorized tallies have a _VEC suffix. The momentum sources are mapped to the smo0_eir_tot, smr0_eir_tot, and smd0_eir_tot arrays in B2.5, respectively. The long-term averages of the radial and diamagnetic momentum components smr_mean and smd_mean and their associated error fields e_smr and e_smd have been added to the *b2faver[ie]* files.

5.2. Full wall contour loading diagnostics:

The **b2plot** option wlld, if given an argument with a nonzero tens digit, will produce the *wlld.dat* file, which contains wall loading information for all elements forming the shadowing structure contour surrounding the plasma.

5.3. New tracing files:

Two new time tracing files *blnn_SPb.trc* and *user_SPb.trc* can be requested from the *b2.user.parameters* namelist, containing particle balance and wall loading information, respectively. The latter is written out by the new *b2mod_userspb* module.

5.4. DivGeo grid visualization:

It is now possible to toggle on or off the emphasis in magenta by **DivGeo** of irregular grid cells (i.e. quadrangular and concave). Note that triangular cells are no longer flagged as being irregular.

5.5. Uinp case build-up stencil files:

The *b2.neutrals.parameters.stencil* file built by **Uinp** now also provides for the settings of the `TARGSP` variables that indicate which species are produced by sputtering from the targets, in agreement with the settings in the **Eirene** input file.

6. Getting a case restarted with v. 3.0.9:

As a consequence of all the changes above, in order to take full advantage of the new 3.0.9 features, users wishing to continue their older runs should keep in mind the issues listed below. If running an existing 3.0.8 case, continuation of this case with the 3.0.9 version should be possible without any special additional preparations, although it is recommended to correct the input files following the information given above or run a **b2yt** conversion step to apply all automated corrections. Because of the various code changes and bug fixes implemented in this new version, users should expect some slight changes to their solutions. Of course, to facilitate converting older cases or building new cases from scratch, the **Uinp**, **b2yt**, and **b2sxdr** programs have been modified to produce input files already compatible with version 3.0.9, which should not preclude a visual inspection of all relevant input files, as is best practice.

1. If you are restarting a case from input files created using code from before November 2019, you should delete the existing *fort.1[0-5]* files as there is an internal change in **Eirene** that has modified the content of these files.
2. If you were previously using NetCDF3, you may not be able to append to your older **.nc* files and would then need to delete/rename them.

7. Additional notes for developers:

7.1. Contributing rules:

A `CONTRIBUTING` file has been added at the top level of the SOLPS-ITER repository (and can also be found as an Appendix to the SOLPS-ITER Manual), which specifies the rules that code contributors are expected to follow when presenting a code change as a Pull Request into the main development branches.

The GIT server now enforces that committed files have Unix-style line endings.

7.2. CI testing protocol:

New Continuous Integration (CI) automatic checks are performed when committing to one of the main code branches or when making a Pull Request. In particular, new tests involve checking against a reference coupled case with OpenMP parallelization, and a dedicated test case to check against restart effects, which proved its value and led to some code changes to eliminate some remaining subtle restart effects in coupled runs. Pull Requests can only be accepted if all CI tests are passed. In cases in which the code changes provided in the Pull Request would predictably make the reference results of the test cases evolve, then a new version of the reference case files should also be provided as part of the Pull Request.

The CI test scripts for SOLPS-ITER can now be found in the *$SOLPSTOP/scripts* directory. They are *SOLPS-ITER_CI_test*, *SOLPS-ITER_debug_CI_test*, *SOLPS-ITER_MPI_CI_test*, and *SOLPS-ITER_OpenMP_CI_test*. They compare results for the new code commits to reference cases in debug mode (**B2.5** stand-alone and coupled), with MPI parallelization (coupled with **Eirene**), with OpenMP parallelization (**B2.5** stand-alone and coupled), and for SN, DN, and slab topologies. The tests cover D-only and multi-species cases, including a majority helium plasma case. For **Eirene** stand-alone cases, we rely on the extensive set of tests performed on the JuGit CI server.

7.3. Local dependencies files:

It is now possible to add, in each of the *config/* directories of the SOLPS-ITER components, a *dependencies.local* file to provide additional dependencies forcing the *Makefile* to compile a file before another. This is particularly useful when compiling with the `-warn interfaces` flag, to make sure that the called routine is compiled before the calling one, so that its interface is properly declared. For each such new dependency one wants to add, create a line with the format: `${OBJDIR}/<calling_routine>.o: ${OBJDIR}/<called_routine>.o` in the *dependencies.local* file.

Also note that, if the top level *Makefile* is modified, this will trigger a re-linking of all the executables included in the compilation target.

A dependency on building the stand-alone no-graphics **Eirene** executable *eirobjx* has been added when compiling *triang* as it is needed by the script of the same name. The compilation of the `Triang` programs has been streamlined to use links to the needed **Eirene** source code instead of to the object and module files to avoid issues coming from the change to `CMake` compilation for **Eirene** stand-alone.

There is also a new `MAKETAGS` option for the *Makefile*s to provide a local override to the command used to build the *TAGS* files, in case the default one is not supported.

7.4. Batch compilation script:

A new batch compilation script, *itermake*, and its associated submission template file, *SLURM.iter_compile*, for use on the ITER SDCC cluster, are available. Their syntax and usage parallel that of the job submission script *itersubmit*. Available options include chain submission, choice of job duration, name, and queue, and it can accommodate any list of compilation targets. The compilation log output is directed to a *make.log.$COMPILER* file. Code ROs on other clusters should feel free to take inspiration from this script to develop their own for use on their devices.

7.5. Miscellaneous:

The *b2trcc* routine from *b2mod_trace* has been renamed *b2trccc* to avoid a clash with one of the Grad-Zhdanov closure routines from the transport side. The *b2mod_average* module has been renamed *b2mod_running_average* and moved from the *src/user* to the *src/modules* directory. The *b2mod_ilutern* module has been converted into the *ilutern_solps* routine.

A new *b2mod_ad* module has been created. This is a dummy module for standard runs, but necessary when using algorithmic differentiation tools to be able to "run the code backwards".

A large number of **Eirene** routines have been modularized, to avoid the use of Fortran `entry` statements. Also, code sections that would have been duplicated because of the dual support of ASCII and JSON-formatted input files have been moved to stand-alone routines, while other routines have been duplicated to handle each format style. The *ioflush* routine has been removed and replaced by its equivalent Fortran intrinsic.

The **Eirene** variables containing the tallies for interactions between the plasma background species were renamed from using the `cppv` string in their names to `mppl`. This also applies to the balance variables saved in the *balance.nc* file and associated Matlab scripts. The `lkind[m|i|p]` arrays have been moved from *eirmod_extrab25* to *eirmod_comusr*.

The *ibase* statement function has been moved to the *b2mod_math* module as *ibase* or *ibase1*, depending on the minimal value returned (`0` or `1`). A new utility function *newunit* mirrors the intrinsic Fortran 2003 function if compiling with an older compiler (i.e. without the `-DF2003` pre-processor pragma).

To avoid confusion with Fortran intrinsic functions, the use of the *lnblnk* function has been replaced by `len_trim` or `trim`. Similarly, the *floor* function, when manipulating single-precision reals, has been renamed to *sfloor*, while the *samax* routine, which handles double-precisions arrays, has been renamed *damax*.